Full Mathematics Inventory

Every equation derived across all pipeline layers

Eton iGEM 2026, Dry Lab | Layers 0, 3, 8, 9, 10, 11, 13, 15 | ★FLAG = unconfirmed parameter ✓ = confirmed from cited paper

I. Layer 0: Wound Biology Cascade

`models/00_wound_biology/wound_biology.py`

State vector

Nine species (all mol L^{-1}): pro-MMP-9, free MMP-9, free TIMP-1, MMP9:TIMP1 complex, free HNE, free A1AT, HNE:A1AT, free SLPI, HNE:SLPI. MMP-3 and plasmin are fixed constants (large excess over pro-MMP-9).

Activation of pro-MMP-9

$$\frac{d[\text{proMMP9}]}{dt} = -\underbrace{\left(\frac{k_{cat}}{K_m}\right)_{\text{MMP3}} C_{\text{MMP3}}}_{\text{MMP3 activation}}[\text{proMMP9}] - \underbrace{\left(\frac{k_{cat}}{K_m}\right)_{\text{plas}} C_{\text{plas}}}_{\text{plasmin activation}}[\text{proMMP9}] \tag{I.1}$$

MMP-9 / TIMP-1 binding

$$\frac{d[\text{MMP9}]}{dt} = +v_{\text{act}} - k_{\text{on,T1}}[\text{MMP9}][\text{TIMP1}] + k_{\text{off,T1}}[\text{MMP9:T1}] \tag{I.2}$$

$$\frac{d[\text{TIMP1}]}{dt} = -k_{\text{on,T1}}[\text{MMP9}][\text{TIMP1}] + k_{\text{off,T1}}[\text{MMP9:T1}] \tag{I.3}$$

$$\frac{d[\text{MMP9:T1}]}{dt} = +k_{\text{on,T1}}[\text{MMP9}][\text{TIMP1}] - k_{\text{off,T1}}[\text{MMP9:T1}] \tag{I.4}$$

HNE inhibition by A1AT (near-irreversible) and SLPI (reversible)

$$\frac{d[\text{HNE}]}{dt} = -k_{\text{on,A1AT}}[\text{HNE}][\text{A1AT}] - k_{\text{on,SLPI}}[\text{HNE}][\text{SLPI}] + k_{\text{off,SLPI}}[\text{HNE:SLPI}] \tag{I.5}$$

$$\frac{d[\text{A1AT}]}{dt} = -k_{\text{on,A1AT}}[\text{HNE}][\text{A1AT}] \tag{I.6}$$

$$\frac{d[\text{HNE:A1AT}]}{dt} = +k_{\text{on,A1AT}}[\text{HNE}][\text{A1AT}] \tag{I.7}$$

(SLPI equations symmetric with $k_{\text{on/off,SLPI}}$.)

HNE half-life

At equilibrium $k_{\text{off,A1AT}} \approx 0$, so the A1AT pseudo-first-order inhibition gives:

$$t_{1/2,\text{HNE}} = \frac{\ln 2}{k_{\text{on,A1AT}}[\text{A1AT}]} = \frac{0.693}{6.5 \times 10^7 \times 1.9 \times 10^{-6}} = 5.5 \text{ ms} \tag{I.8}$$

Key parameters

Parameter	Value	Units	Source	Status
$k_{\text{on,TIMP1}}$	2.48×10^5	$M^{-1}s^{-1}$	Olson 1997	✓
$k_{\text{off,TIMP1}}$	2.1×10^{-3}	s^{-1}	Olson 1997	✓
$k_{\text{on,A1AT}}$	6.5×10^7	$M^{-1}s^{-1}$	Beatty 1980	✓
$k_{\text{off,A1AT}}$	10^{-6}	s^{-1}	placeholder	★FLAG
$k_{\text{on,SLPI}}$	10^5	$M^{-1}s^{-1}$		★FLAG
$(k_{cat}/K_m)_{\text{MMP3}}$	5×10^3	$M^{-1}s^{-1}$		★FLAG

II. Layer 3: Mass Transport and Fouling

`models/03_mass_transport/`

3a. Enzyme diffusion to electrode (1D PDE + Robin BC)
Enzyme concentration $[E](z,t)$ in a channel of height $h = 100\,\mu$m:

$$\frac{\partial [E]}{\partial t} = D_E \frac{\partial^2 [E]}{\partial z^2}, \quad 0 < z < h \tag{II.1}$$

$$\left.\frac{\partial [E]}{\partial z}\right|_{z=h} = 0 \qquad \text{(no flux at lid)} \tag{II.2}$$

$$D_E \left.\frac{\partial [E]}{\partial z}\right|_{z=0} = \kappa(t)\,[E](0,t) \qquad \text{(Robin sink at electrode)} \tag{II.3}$$

$$\frac{d\Gamma}{dt} = -\kappa(t)\,[E](0,t), \qquad \kappa(t) = \frac{k_{cat}}{K_{m,\text{surf}}}\,\Gamma(t) \tag{II.4}$$

Damköhler number: $\text{Da} = \kappa(0)h/D_E \approx 0.03 \ll 1$ (reaction-limited, diffusion not rate-limiting).

Stokes-Einstein diffusivity for MMP-9 (MW = 92 kDa, $r_h = 5.0$ nm):

$$D_E = \frac{k_B T}{6\pi\eta r_h} = \frac{1.38\times10^{-23}\times 298}{6\pi\times 8.9\times10^{-4}\times 5.0\times10^{-9}} = 4.9\times10^{-11}\ \text{m}^2\text{s}^{-1} \tag{II.5}$$

3b. Three-species inhibitor reaction-diffusion

$$\frac{\partial [E]}{\partial t} = D_E\frac{\partial^2 [E]}{\partial z^2} - k_{\text{on}}[E][I] \tag{II.6}$$

$$\frac{\partial [I]}{\partial t} = D_I\frac{\partial^2 [I]}{\partial z^2} - k_{\text{on}}[E][I] \tag{II.7}$$

$$\frac{\partial [C]}{\partial t} = D_C\frac{\partial^2 [C]}{\partial z^2} + k_{\text{on}}[E][I] \tag{II.8}$$

$[I]$ and $[C]$ have no-flux BCs at both walls; only $[E]$ carries the Robin surface sink.

3c. SAM fouling (Langmuir kinetics)
Fractional fouling coverage $\Theta_f \in [0,1]$:

$$\frac{d\Theta_f}{dt} = k_f[P]_{\text{bulk}}(1-\Theta_f) - k_d\Theta_f \tag{II.9}$$

Steady state: $\Theta_f^* = k_f[P]/(k_f[P]+k_d)$. Effect on cleavage fraction:

$$F_k^{\text{fouled}}(T) = 1 - \exp(-k_{\text{obs}}\,\bar{\alpha}\,T)\,, \qquad \bar{\alpha} = \frac{1}{T}\int_0^T \frac{h}{h + h_{f,\max}\Theta_f(t)}\,dt \tag{II.10}$$

3d. In-device inhibitor binding (well-mixed)
MMP-9/TIMP-1 (reversible):

$$\frac{d[\text{MMP9}]}{dt} = \frac{d[\text{TIMP1}]}{dt} = -k_{\text{on}}[\text{MMP9}][\text{TIMP1}] + k_{\text{off}}[\text{MMP9:T1}] \tag{II.11}$$

HNE/A1AT (irreversible):

$$\frac{d[\text{HNE}]}{dt} = \frac{d[\text{A1AT}]}{dt} = -k_{\text{on,A1AT}}[\text{HNE}][\text{A1AT}] \tag{II.12}$$

III. Layer 8: Probe Tether Geometry (FJC)

`models/swv_forward_model/swv_forward_model.py`

PEG6 spacer: Flory radius (hard wall)
PEG6 has $N_{\text{EO}} = 6$ ethylene oxide units, each of length $l_{\text{EO}} = 0.35$ nm, with Kuhn length $l_K = 0.76$ nm:

$$N_{K,\text{PEG}} = \frac{N_{\text{EO}}\,l_{\text{EO}}}{l_K} = \frac{6\times 0.35}{0.76} = 2.76 \tag{III.1}$$

In the dilute mushroom regime (good solvent), the Flory exponent is 3/5:

$$d_{\text{PEG}} = R_F = N_{K,\text{PEG}}^{3/5}\,l_K = (2.76)^{0.6}\times 0.76 = 1.398\ \text{nm} \tag{III.2}$$

Peptide tether: freely jointed chain

For SGKIPRTLTA ($N_r = 10$ residues, $l_0 = 0.38$ nm/residue, $b = 0.76$ nm Kuhn length):

$$N_K = \frac{N_r\, l_0}{b} = \frac{10 \times 0.38}{0.76} = 5 \tag{III.3}$$

$$\langle r^2 \rangle = N_K b^2 = 5 \times (0.76)^2 = 2.888 \text{ nm}^2 \tag{III.4}$$

$$\sigma_z = \sqrt{\frac{\langle r^2 \rangle}{3}} = \sqrt{\frac{2.888}{3}} = 0.981 \text{ nm} \tag{III.5}$$

The 1/3 factor comes from isotropy: $\langle \cos^2 \theta \rangle = 1/3$ for a uniformly distributed random walk.

Distance distribution

The MB-to-gold distance $d = d_{\text{PEG}} + z$ follows a half-Gaussian (positive half of a zero-mean Gaussian with std σ_z):

$$\boxed{P(d) = \sqrt{\frac{2}{\pi}} \frac{1}{\sigma_z} \exp\left(-\frac{(d - d_{\text{PEG}})^2}{2\sigma_z^2} \right), \quad d \geq d_{\text{PEG}}} \tag{III.6}$$

Mean distance:

$$\langle d \rangle = d_{\text{PEG}} + \sigma_z \sqrt{\frac{2}{\pi}} = 14.0 + 9.81 \times 0.798 = 21.8 \text{ Å} \tag{III.7}$$

IV. Layer 10: Electron Transfer Kinetics

`models/swv_forward_model/swv_forward_model.py`

Marcus theory: activation barrier

Free-energy surfaces are parabolas in reaction coordinate q. At overpotential $\eta = E - E^{\circ\prime}$:

$$\Delta G^{\ddagger} = \frac{(\Delta G^0 + \lambda)^2}{4\lambda} \xrightarrow{\Delta G^0 = 0} \frac{\lambda}{4} \tag{IV.1}$$

Reorganisation energy: $\lambda = 0.85$ eV ✓ (Chidsey 1991).

Butler-Volmer as the Marcus low-η limit

First-order expansion of $\Delta G^{\ddagger}$ in η:

$$\Delta G^{\ddagger}(\eta) \approx \frac{\lambda}{4} - \frac{\alpha n_e F \eta}{RT} \tag{IV.2}$$

Transfer coefficient from Marcus geometry:

$$\alpha = \frac{1}{2}\left(1 + \frac{n_e F \eta}{\lambda}\right) \xrightarrow{\eta \to 0} 0.5 \quad \text{(measured: } \alpha = 0.37 \text{ for MB, ✓ DD2017)} \tag{IV.3}$$

Rate constants:

$$k_f(E) = k^0 \exp\left(-\frac{\alpha n_e F (E - E^{\circ\prime})}{RT} \right) \tag{IV.4}$$

$$k_b(E) = k^0 \exp\left(+\frac{(1-\alpha) n_e F (E - E^{\circ\prime})}{RT} \right) \tag{IV.5}$$

Tunnelling decay and distance dependence

Wavefunction amplitude through a barrier of height V_0:

$$\psi(d) \propto e^{-\kappa_{\text{tun}} d}, \qquad \kappa_{\text{tun}} = \frac{\sqrt{2 m_e V_0}}{\hbar} \tag{IV.6}$$

Rate scales as $|\psi|^2$:

$$k^0(d) = k_0^0\, e^{-\beta d}, \qquad \beta = \frac{2\sqrt{2 m_e V_0}}{\hbar} \tag{IV.7}$$

From Smalley (1995): $\beta = 1.21/1.27$ Å $= 0.95$ Å^{-1} for alkanethiol ✓; value for PEG bridge ★ FLAG.

Reference rate at PEG6 tip

$$k_{\text{ref}}^0 = k_0^0\, e^{-\beta d_{\text{PEG}}} = 1.5 \times 10^8 \times e^{-0.95 \times 1.40} \approx 127 \text{ s}^{-1} \tag{IV.8}$$

(★ FLAG on k_{ref}^0: β_{PEG} not measured.) Cross-check: Chidsey (1991) reports $k^0 = 1.3$ s^{-1} for Fc-C16 at $d = 20.3$ Å, back-calculating $k_0^0 = 1.3\, e^{0.95 \times 2.03} \approx 2.7 \times 10^8$ s^{-1}, consistent with DD2017 value of 1.5×10^8 s^{-1} ✓.

Tafel plot slopes

From the BV expressions:

$$\frac{\partial \ln k_f}{\partial E} = -\frac{\alpha n_e F}{RT}, \qquad \frac{\partial \ln k_b}{\partial E} = +\frac{(1-\alpha) n_e F}{RT} \tag{IV.9}$$

At $T = 298$ K, $n_e = 2$, $\alpha = 0.37$: $\text{slope}_f = -0.37 \times 2 \times 38.9 = -28.8$ V^{-1}; $\text{slope}_b = +0.63 \times 2 \times 38.9 = +49.0$ V^{-1}.

V. Layer 9: Three-State Surface Cleavage ODE

`models/09_surface_kinetics/surface_kinetics.py`

Mechanism: $E + \Gamma_S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \Gamma_{ES} \xrightarrow{k_{cat}} E + \Gamma_P$.

$$\frac{d\Gamma_S}{dt} = k_{-1}\Gamma_{ES} - k_1[E]\Gamma_S \tag{V.1}$$

$$\frac{d\Gamma_{ES}}{dt} = k_1[E]\Gamma_S - (k_{-1} + k_{cat})\Gamma_{ES} \tag{V.2}$$

$$\frac{d\Gamma_P}{dt} = k_{cat}\Gamma_{ES} \tag{V.3}$$

Conservation: $\Gamma_S + \Gamma_{ES} + \Gamma_P = \Gamma_0 = 3 \times 10^{-12}$ mol cm^{-2} ✓.

Analytic pseudo-first-order limit (valid when $k_1[E] \ll k_{cat}$, confirmed at clinical concentrations):

$$\boxed{F_k(T) = 1 - e^{-k_{\text{eff,surf}}[E]T}} \tag{V.4}$$

where $k_{\text{eff,surf}} = \eta \times (k_{cat}/K_m)_{\text{sol}}$. From Kridel (2001): $(k_{cat}/K_m)_{\text{sol}} = 67{,}500$ M^{-1}s^{-1} ✓. Surface efficiency $\eta = 0.20$ (central estimate) ★FLAG.

VI. Layer 11/13: SWV Forward Model

`models/swv_forward_model/swv_forward_model.py`

Butler-Volmer surface ODE

MB is surface-confined (no diffusion). One ODE per half-pulse:

$$\frac{d\Gamma_{\text{red}}}{dt} = k_f\Gamma_{\text{tot}} - (k_f + k_b)\Gamma_{\text{red}} \tag{VI.1}$$

Exact analytic solution (5-step integrating factor)

Step 1. Standard form: $\dot{\Gamma}_{\text{red}} + b\Gamma_{\text{red}} = a$, where $a = k_f\Gamma_{\text{tot}}$, $b = k_f + k_b$.

Step 2. Multiply by integrating factor $\mu = e^{bt}$: $e^{bt}\dot{\Gamma}_{\text{red}} + be^{bt}\Gamma_{\text{red}} = ae^{bt}$.

Step 3. Recognise as product rule: $\frac{d}{dt}[\Gamma_{\text{red}}e^{bt}] = ae^{bt}$.

Step 4. Integrate from 0 to τ: $\Gamma_{\text{red}}(\tau)e^{b\tau} - \Gamma_{\text{red}}(0) = (a/b)(e^{b\tau} - 1)$.

Step 5. Divide by $e^{b\tau}$, substitute $a/b = \Gamma_{\text{eq}}$:

$$\boxed{\Gamma_{\text{red}}(\tau) = \Gamma_{\text{eq}} + \left[\Gamma_{\text{red}}(0) - \Gamma_{\text{eq}}\right] e^{-(k_f+k_b)\tau}} \tag{VI.2}$$

where $\Gamma_{\text{eq}} = k_f/(k_f + k_b)\,\Gamma_{\text{tot}}$ and $\tau = 1/(2f) = 33.3$ ms.

Dimensionless kinetic parameter

$$\kappa = k^0/f_{\text{SWV}} \tag{VI.3}$$

Three regimes: $\kappa \ll 1$ (irreversible, $\Delta I \to 0$), $\kappa \approx 1$ (quasi-reversible, ΔI maximised), $\kappa \gg 1$ (reversible, $\Delta I \to 0$). Operating point: $\kappa_{\text{tip}} = 127/15 = 4.2$ ★FLAG.

Sampled currents and net SWV difference

Current sampled at end of half-pulse (anodic positive):

$$I(E) = n_e F A\left[k_b\Gamma_{\text{red}}(\tau) - k_f\Gamma_{\text{ox}}(\tau)\right] \tag{VI.4}$$

Net SWV current at staircase step E_{step}:

$$\Delta I(E_{\text{step}}) = I_f(E_{\text{step}} + E_{\text{sw}}) - I_b(E_{\text{step}} - E_{\text{sw}}) \tag{VI.5}$$

Distance-averaged voltammogram

$$\Delta I(E; F_k) = (1 - F_k)\,\Gamma_{\text{tot}}\,A \int_{d_{\text{PEG}}}^{d_{\max}} P(d)\,\Delta I_{\text{norm}}(E;\, k^0(d))\,dd \tag{VI.6}$$

Computed by 80-point quadrature. Probes near d_{PEG} dominate because $P(d)$ peaks at the hard wall and $k^0(d)$ falls exponentially with d.

SWV charging-current cancellation

Double-layer charging current decays on RC timescale $\tau_{\text{RC}} \sim 1$ ms. Faradaic current persists on $1/(k_f + k_b) \sim 33$ ms. Sampling at the end of the half-pulse eliminates I_c. Subtracting $I_f - I_b$ removes residual symmetric background. Net SNR $\approx 8\times$ better than CV.

VII. Layer 13: Laviron Limit and FWHM

`models/swv_forward_model/swv_forward_model.py`

Reversible peak shape (Laviron 1979)

In the limit $E_{\text{sw}} \to 0$, SWV converges to:

$$|\Psi| = \frac{\eta_L}{(1 + \eta_L)^2}, \qquad \eta_L = \exp\left[\frac{n_e F(E - E^{\circ\prime})}{RT}\right] \tag{VII.1}$$

Peak maximum at $E = E^{\circ\prime}$: $|\Psi|_{\max} = 1/4$.

Analytic FWHM derivation

Set $|\Psi| = |\Psi|_{\max}/2 = 1/8$:

$$\frac{\eta_L}{(1 + \eta_L)^2} = \frac{1}{8} \;\Rightarrow\; \eta_L = 3 \pm 2\sqrt{2} \tag{VII.2}$$

The two half-maximum points are at $E_\pm = (RT/n_e F)\ln(3 \pm 2\sqrt{2})$.

$$\text{FWHM} = E_+ - E_- = \frac{2RT}{n_e F}\ln(3 + 2\sqrt{2}) = \frac{3.53\,RT}{n_e F} \tag{VII.3}$$

For $n_e = 2$ at $T = 298$ K: FWHM = 45.3 mV ✓ (Laviron 1979).

VIII. Layer 15: Calibration and Limit of Detection

`models/15_calibration/calibration.py`

Ferrocene ratiometric correction

Fc is always reversible ($\kappa_{\text{Fc}} = 9000 \gg 1$, ✓):

$$F_k^{\text{rec}} = 1 - \frac{A_{\text{MB}}(T)/A_{\text{Fc}}}{A_{\text{MB}}(0)/A_{\text{Fc}}} \tag{VIII.1}$$

Cancels ±15% electrode-to-electrode drift in Γ_{tot}. Residual noise: $\sigma_{F_k} = 3.4\%$.

Calibration linearity

From the distance-averaging integral, $A_{\text{area}}(F_k) \propto (1 - F_k)$ to $< 2\%$ nonlinearity over $F_k \in [0, 0.95]$:

$$A_{\text{area}}(F_k) = A_{\text{area}}(0)\,(1 - F_k) \tag{VIII.2}$$

Limit of detection

At 3σ criterion:

$$[\text{MMP-9}]_{\text{LOD}} = \frac{3\sigma_{F_k}}{k_{\text{eff,surf}}\,T} = \frac{3 \times 0.034}{\eta \times 67{,}500 \times T} \tag{VIII.3}$$

η	k_{eff} ($\text{M}^{-1}\text{s}^{-1}$)	LOD (30 min)	LOD (60 min)	Below 6 nM?
0.05	3,375	17 nM	8.5 nM	No / Yes
0.20	**13,500**	**4.3 nM**	**2.1 nM**	**Yes**
0.50	33,750	1.7 nM	0.9 nM	Yes

Temperature corrections

MB formal potential: $\partial E^{\circ\prime}/\partial T = -1.5$ mV K^{-1} (DD2017 ✓). From lab (25 °C) to wound (37 °C): $\Delta E^{\circ\prime} = -18$ mV.

k^0_{ref} via Arrhenius ($E_a = 0.30$ eV, inferred from Chidsey 1991):

$$k^0(T) = k^0_{\text{ref}} \exp\left[-\frac{E_a}{k_B}\left(\frac{1}{T} - \frac{1}{298}\right)\right] \tag{VIII.4}$$

From 25 °C to 37 °C: k^0 increases $\approx 70\%$, $\kappa_{\text{tip}} : 4.2 \to 7.1$.

System Summary

Layer	Model	States	Type	Solver / solution
L0	Wound biology cascade	9 ODEs	nonlinear	`solve_ivp` RK45
L3a	Enzyme diffusion + depletion	PDE + 1 ODE	linear PDE	Crank-Nicolson FD
L3b	3-species react.-diffusion	3 PDEs	nonlinear PDE	op.-split FD
L3c	SAM fouling (Langmuir)	1 ODE	nonlinear	`solve_ivp` RK45
L3d	In-device inhibition	2×3 ODEs	nonlinear	`solve_ivp` Radau
L8	FJC tether + $P(d)$	algebraic	statistical	closed form
L9	3-state surface MM	3 ODEs	nonlinear	RK45 / analytic limit
L10	Marcus / BV / tunnelling	algebraic		closed form
L11/13	BV surface ODE (SWV)	1 per half-pulse	linear ODE	**exact analytic**
L13	Laviron FWHM	algebraic		analytic ($3.53\,RT/n_eF$)
L15	LOD / ratiometric	algebraic		closed form